HEATTrack: History Evidence Injection and Target-Aware Prompting for Aerial Multispectral Single-Object Tracking

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***Abstract*—Aerial multispectral single-object tracking (SOT) is fundamental to inspection, search-and-rescue, and long-range surveillance from moving platforms. However, it remains highly vulnerable to temporal drift caused by tiny targets, fast platform motion, abrupt viewpoint changes, and cluttered backgrounds. Multispectral imagery (MSI) provides complementary reflectance cues beyond RGB, but reliably exploiting spectral information under band noise and background variability is still challenging. We present HEATTrack, a Transformer-based framework that balances short-term temporal evidence and long-term target identity cues for stable aerial multispectral tracking. First, history evidence injection (HEI) injects lightweight short-term temporal evidence by rasterizing recent predictions into boxmaps and fusing them with selected-band historical observations, yielding evidence maps that suppress irrelevant background and provide observation-conditioned guidance under noisy historical predictions. The history evidence module is trained with a perturbation-and-annealing strategy to reduce the mismatch between clean training histories and imperfect inference-time predictions. Second, target-aware multitemplate prompting (TAMTP) preserves anchor-guided target identity cues by injecting compact structural and spectral signatures into multitemplate prompts and aggregating templates with reliability-aware weights, thereby strengthening identity discrimination under distractors. Extensive experiments on MUST and MSITrack demonstrate consistent gains over competitive multispectral and general-purpose baselines. On MUST, HEATTrack improves the area under curve (AUC) from 62.8 to 67.1 over our reproduced UNTrack* baseline. On MSITrack, HEATTrack achieves 56.2 AUC, improving over our reproduced UNTrack* by 5.7 points. The source code, trained models, and reproduction files will be released upon publication.**

***Index Terms*—Aerial multispectral tracking, short-term temporal evidence, single-object tracking (SOT), target-aware prompting.**

I. Introduction

AERIAL single-object tracking (SOT) from moving platforms serves as a cornerstone capability for uncrewed aerial vehicles (UAVs), helicopters, and fixed-wing aircraft, supporting autonomous missions such as inspection, search-and-rescue, and persistent surveillance. In recent years, SOT systems have evolved from Siamese matching paradigms [1], [2] and mask-based trackers [3] to discriminative learning and Transformer-based frameworks [4], [5], substantially advancing general-purpose tracking performance. Yet aerial tracking remains particularly challenging. Unlike ground-view scenarios, aerial videos captured from moving platforms often exhibit rapid ego-motion, abrupt viewpoint changes, and severe scale variation. Under such conditions, targets usually occupy only a small number of pixels and are easily overwhelmed by cluttered backgrounds [6]. These factors make aerial tracking highly susceptible to *temporal drift*: even minor localization errors can be amplified through the crop-update feedback loop, progressively contaminating the target representation and eventually causing irreversible failure. Consequently, recent studies on aerial tracking have increasingly emphasized motion-aware temporal reasoning and stability, rather than relying solely on appearance matching [6], [7], [8].

To alleviate these aerial-view ambiguities, multispectral imagery (MSI) provides complementary reflectance cues beyond RGB texture. In aerial scenes, spectral responses can remain more stable under illumination changes and offer discriminative target–background evidence when spatial appearance becomes weak or ambiguous. For multispectral/hyperspectral object tracking, a straightforward strategy is to adapt strong RGB trackers through band regrouping (false-color) or explicit band selection [9], [10], [11], channel inflation with parameter transfer [12], or multiband early fusion with spectral–spatial feature extraction [13], [14], [15]. While these adaptations improve cross-band compatibility, they often treat spectral channels as passive inputs and do not explicitly model when spectral cues are reliable. As a result, they remain vulnerable to sensor misalignment, band noise, and background spectral variability encountered in practice.

More recent multispectral/hyperspectral trackers have begun to exploit spectral cues through spectrum-aware fusion [16], [17], memory-based conditioning [18], [19], and prompt-style spectral prior injection [20], [21], [22], [23]. In parallel, prompt learning has also emerged in generic RGB tracking as an efficient mechanism for query adaptation and temporal conditioning [24], [25], [26], [27], [28], [29]. However, existing methods typically treat history and spectral priors as static cues

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rather than dynamic evidence whose reliability should depend on current observations. This limitation becomes especially severe in aerial scenarios, where platform motion, viewpoint instability, and tiny-target ambiguity can quickly turn stale history or noisy bands into misleading guidance. The recent release of benchmarks such as MUST [30] and MSITrack [31] has begun to standardize evaluation for multispectral SOT. In particular, MUST provides a dedicated aerial benchmark under moving-platform disturbances, while MSITrack offers a complementary test bed for evaluating performance in broader multispectral tracking scenarios.

Despite this progress, two critical gaps remain in aerial multispectral SOT under moving-platform conditions. First, *short-term temporal adaptability* is needed to exploit recent motion and localization cues without over-trusting noisy historical predictions, which are a major source of drift in aerial tracking. Second, *long-term identity stability* is needed to preserve target-specific appearance, structural, and spectral cues when recent observations become ambiguous or contaminated by distractors. The key challenge is therefore not simply to add more temporal or spectral inputs, but to use short-term evidence and long-term identity cues in a reliability-aware manner.

In this work, we address these challenges with **HEATTrack**, a tracking framework that combines history evidence injection (HEI) and target-aware multitemplate prompting (TAMTP) on top of the UNTrack baseline [30]. Rather than redesigning the underlying tracking architecture, HEATTrack strengthens aerial multispectral tracking by balancing two complementary sources of robustness: short-term temporal evidence from recent frames and long-term identity cues from anchor-guided templates. First, HEI aggregates recent history into explicit *evidence maps*, which provide observation-conditioned spatial priors derived from both predicted states and selected-band historical observations. Unlike implicit memory tokens, these maps preserve temporally useful localization cues while reducing over-reliance on noisy geometric history. Second, TAMTP injects compact structural and spectral signatures into the template representation and fuses multiple templates with reliability-aware weights, thereby preserving stable target identity cues under distractors.

To reduce the train–test mismatch caused by overly clean history during training, we further develop a perturbation-and-annealing scheme that progressively exposes the tracker to realistic history noise. Extensive experiments on MUST, our primary aerial multispectral benchmark, show that HEATTrack significantly improves robustness under moving-platform disturbances, raising area under curve (AUC) from 62.8 to 67.1 (+4.3) and normalized precision from 77.5 to 83.6 (+6.1) over our reproduced UNTrack* baseline. On MSITrack, a complementary benchmark for broader multispectral tracking scenarios, HEATTrack achieves 56.2 AUC, improving over our reproduced UNTrack* baseline (50.5 AUC) by 5.7 points and showing consistent gains across major challenge attributes.

Our main contributions are threefold.

1) We propose a short-term HEI mechanism for aerial multispectral SOT. HEI transforms recent predicted states and selected-band historical observations into explicit spatial evidence maps, providing lightweight temporal guidance while reducing over-reliance on noisy geometric history. A perturbation-and-annealing training strategy is further used as a noise-aware training component to reduce the mismatch between clean training histories and imperfect inference-time predictions.
2) We develop an anchor-guided TAMTP mechanism for long-term identity preservation. TAMTP encodes compact appearance, structural, and spectral target signatures, and aggregates multiple template prompts with reliability-aware weights. To handle noisy auxiliary templates, it introduces an anchor-biased fallback to the first-frame template, enabling stable identity prompting under tracking noise, box misalignment, and distractor contamination.
3) We provide a comprehensive evaluation on MUST and MSITrack, together with targeted analyses that examine input-protocol fairness, noisy-history robustness, evidence-map interpretability, history-length behavior, and template-fusion reliability. These results validate the complementary roles of short-term temporal evidence and long-term identity cues in improving aerial multispectral tracking stability and accuracy.

II. Related Work

A. Aerial SOT

Aerial SOT is typically performed under rapid ego-motion, abrupt viewpoint changes, severe scale variation, frequent occlusion, and extremely small targets, making it more vulnerable to temporal drift and error accumulation than many ground-view settings. Similar scale and clutter challenges have also motivated multiscale fusion designs in remote sensing object detection [32], whereas aerial SOT additionally requires maintaining a specific target identity over time. To support research in this area, the community has established a series of aerial benchmarks and protocols, including UAV123/UAV20L [33], DTB70 [34], VTUAV [35], and large-scale webly collected resources such as WebUAV-3M [36]. Recent surveys further summarize the benchmarks, challenges, and methodological trends of aerial tracking, highlighting the growing importance of exploiting temporal context beyond per-frame appearance matching [6].

Modern SOT methods can be broadly categorized into Siamese matching, discriminative learning with online adaptation, and Transformer-based paradigms. Siamese trackers, such as SiamFC, SiamRPN, and SiamMask [1], [2], [37], learn target-search similarity and localize the target via response matching or proposal regression. Discriminative trackers, represented by ATOM, DiMP, and PrDiMP [3], [38], [39], emphasize target–background separation and online adaptation, improving robustness to distractors at the cost of more complex update dynamics. More recently, Transformer-based trackers such as TransT, STARK, MixFormer, and OSTrack [4], [40], [41], [42] replace hand-crafted matching with attention-based target-search interaction and direct box prediction, substantially advancing generic tracking performance.

A line of work particularly relevant to aerial tracking is explicit temporal modeling, which aims to reduce drift by

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exploiting multiframe context, motion regularities, and history-dependent state priors. For aerial scenarios, TCTrack [8] and SmallTrack [7] incorporate temporal/contextual cues and frequency-domain priors tailored to small targets and moving-platform motion patterns. More generally, recent trackers increasingly leverage temporal cues through token propagation, memory, or query evolution. For example, ODTrack [24] learns online dense temporal tokens to capture short-term dynamics efficiently, while AQATrack [26] explores adaptive query assignment and temporal query evolution for robust regression under appearance changes. Autoregressive formulations, such as SeqTrack and ARTrack/ARTrackV2 [43], [44], [45], further recast tracking as sequence modeling, where the current prediction is explicitly conditioned on previous states.

Prompt-based conditioning has also emerged as an efficient means of injecting temporal or motion priors and suppressing distractors. Representative designs include explicit visual prompting and spatiotemporal conditioning (EVPTrack [25], ACTrack [46]), foundation-model-guided prompting (PiVOT [47]), motion-prompt modules and plug-in enhancement blocks [5], [48], as well as efficiency-oriented zooming and local-neighborhood reasoning (ZoomTrack [27], NeighborTrack [28]).

Overall, existing aerial trackers increasingly rely on temporal context, memory, or prompt conditioning to mitigate drift. These designs are effective, but most of them propagate history through latent tokens, evolving queries, or updated templates, making it difficult to explicitly inspect or regulate how noisy intermediate predictions influence the next localization step. HEATTrack instead represents recent tracking history as short-term spatial evidence maps constructed from predicted states and historical observations, making temporal guidance more explicit and reducing over-reliance on noisy geometric priors.

B. Multispectral and Hyperspectral Tracking

Beyond RGB imagery, multispectral and hyperspectral sensing provide material-sensitive cues that can improve target discrimination under low illumination, camouflage, or weak texture. At the same time, they introduce additional challenges, including modality gaps, band variability, sensor noise, misalignment, and limited annotated data. Recent RGB-X surveys emphasize the need for unified modeling and modality-adaptive mechanisms in such settings [49].

For general RGB-X tracking, prompt-style adaptation is widely adopted to transfer knowledge from strong RGB backbones to other modalities. For example, ViPT [50] learns modality-relevant prompts on top of a frozen tracker to improve parameter efficiency and transferability. Unified modeling across modalities has also been explored through sequence-to-sequence or single-model formulations, such as SeqTrackv2 [51], Un-Track [52], and efficient cross-modal interaction frameworks like EMTrack [53]. For aerial multispectral tracking in particular, MUST establishes a dedicated benchmark under moving-platform disturbances, and UNTrack [30] provides a strong Transformer-based baseline under this protocol. MSITrack [31] further offers a complementary multispectral SOT benchmark beyond UAV-specific aerial scenes.

Hyperspectral tracking has advanced from early spectral–spatial fusion to more recent Transformer-based interaction, band modeling, and robustness-oriented reasoning. A recent survey [54] summarizes representative paradigms such as Siamese matching, attention-based band weighting, and Transformer fusion, and highlights the importance of spectral priors for suppressing distractors. Representative methods for band modeling and importance estimation include SiamBAG, SEE-Net, and spectral-angle-aware fusion [10], [11], [14], [55], [56], while transformer-based spectral–spatial interaction and efficient fusion frameworks further improve robustness [15], [57], [58]. To address band differences and domain shifts, DA-aware Transformers explicitly model cross-domain generalization [59], and complementary material-/motion-aware reasoning as well as long-sequence template mechanisms have also been explored [23], [60], [61].

Prompt learning and feature injection are particularly influential in multispectral and hyperspectral tracking, since they provide a lightweight way to inject target-specific spectral priors under limited supervision. Representative examples include visual prompting and modality prompters (VP-HOT [62], PHTrack [19]), spectral-similarity prompts and dual-stream prompt designs [20], [21], unified spectral–spatial–temporal conditioning (SSTtrack [16]), frequency-aware filtering with template-enhanced interaction (ProFiT [18]), and memory-enhanced prompting (SpectralTrack [17], SPHST [63]). History-conditioned prompting for universal hyperspectral tracking (HOPL [22]) is also closely related to our motivation of combining temporal evidence with target-aware cue injection.

Although multispectral and hyperspectral tracking share the common goal of exploiting spectral information, they are not identical settings. We review hyperspectral trackers because they reveal transferable design dimensions, including band modeling, spectral prior injection, and temporal conditioning. In contrast, our focus is aerial multispectral SOT under moving-platform disturbances, where spectral redundancy is lower, platform-induced drift is stronger, and reliable use of short-term history becomes particularly critical.

Taken together, recent multispectral and hyperspectral trackers show that spectral priors, temporal memory, and prompt-style conditioning are all useful for improving target discrimination. However, in aerial multispectral SOT, these cues are also vulnerable to tracking-induced uncertainty. Historical predictions may drift under rapid platform motion, selected bands may contain background leakage, and auxiliary templates may be contaminated by box misalignment or nearby distractors. This motivates us to balance short-term temporal evidence and long-term target identity cues in a reliability-aware manner. HEATTrack externalizes recent history as observation-conditioned evidence maps, rather than keeping it only in latent memory or updated templates. It also injects compact spectral-structural target signatures into anchor-guided multitemplate prompts and fuses them with reliability-aware weights. In this way, the proposed design focuses on making both recent observations and anchor-guided

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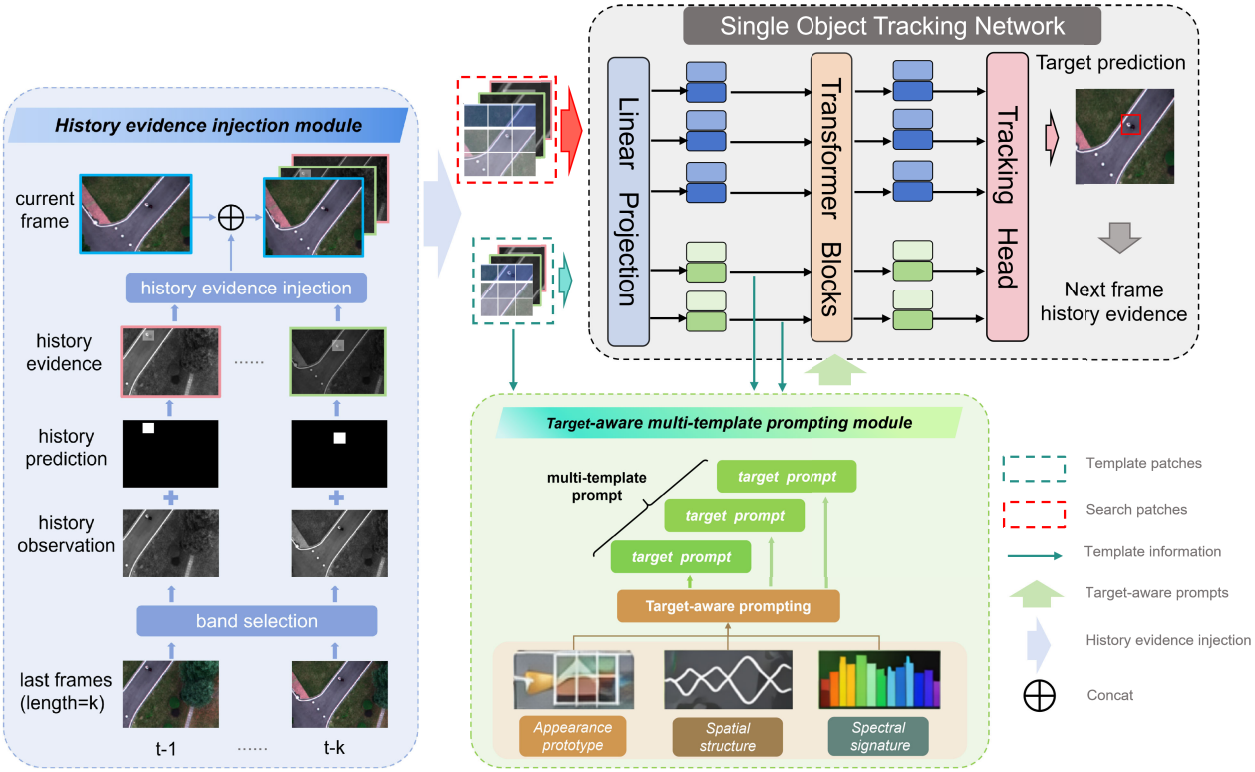


Fig. 1. Overall architecture of HEATTrack. The proposed framework consists of two core modules: the HEI module and the TAMTP module. HEI converts recent predicted states and historical observations into short-term evidence maps, which are concatenated with the current-frame input before the projection layer. TAMTP generates anchor-guided target-aware multitemplate prompts and injects them into the Transformer blocks to preserve stable identity cues under distractors.

templates useful under noise, rather than simply adding more temporal or spectral inputs.

III. Methods

To improve temporal robustness in aerial multispectral SOT, we propose HEATTrack, as illustrated in Fig. 1. HEATTrack is built upon the UNTrack baseline [30] and augments the original pipeline with two complementary components: HEI and TAMTP. Specifically, HEI converts recent predicted states and historical observations into explicit evidence maps that provide lightweight short-term spatial evidence for the current frame, while TAMTP preserves anchor-guided target identity cues through reliability-aware multitemplate prompts. Following UNTrack, the SOT network consists of the input projection, the Transformer backbone, and the tracking head. HEI modifies the input construction and the first projection layer, whereas TAMTP injects target-aware prompts into the Transformer blocks; the core Transformer blocks and tracking head remain unchanged. All temporal cues are constructed strictly from past frames, and train–test mismatch is reduced through a perturbation-and-annealing training strategy.

Different from methods that propagate history mainly through latent tokens, evolving queries, or updated templates, HEATTrack exposes recent history to the tracker as input-level spatial evidence. In parallel, TAMTP uses the first-frame template as a stable identity anchor and fuses auxiliary templates according to their reliability. This design allows short-term evidence and long-term identity cues to complement each other under noisy aerial tracking conditions.

A. HEI Module

In aerial multispectral SOT, tiny targets and heavy background clutter make localization unstable and prone to drift under moving-platform disturbances. HEI is designed to provide lightweight short-term temporal evidence from recent frames without over-trusting noisy historical predictions. Specifically, it decouples two roles of short-term tracking history: predicted boxes provide geometric hypotheses, while selected-band historical observations provide target-related appearance/spectral evidence within these hypotheses. The two sources are fused into explicit evidence maps so that the model can use recent localization cues while retaining observation evidence for suppressing background leakage. Meanwhile, train–test consistency is improved through perturbation-and-annealing training.

1) Construction of Historical Evidence Maps: To make historical predictions usable by the network in a simple yet effective way, we fuse historical observations with historical predicted boxes. Given the current frame I_t, we use the most recent k past frames $\{H_{t-i}\}_{i=1}^{k}$ and their predicted boxes $\{\hat{b}_{t-i}\}_{i=1}^{k}$ to construct evidence maps. For each past frame $t-i$, we first rasterize the predicted box $\hat{b}_{t-i}$ into a binary box map M_{t-i} with the same spatial resolution as the input

$$M_{t-i}(p) = \mathbf{1}\left[p \in \hat{b}_{t-i}\right] \tag{1}$$

where p denotes a pixel location.

We then fuse the prediction mask with the historical observation H_{t-i} to obtain a historical evidence map. Here, H_{t-i}

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denotes the selected-band historical observation from frame $t-i$ rather than the full multispectral frame; by default, each E_{t-i} is therefore a single-channel evidence map

$$E_{t-i} = H_{t-i} \odot (\alpha + (1-\alpha) M_{t-i}) \tag{2}$$

where $\alpha \in [0,1]$ controls background attenuation ($\alpha = 0.5$ by default), and $\odot$ denotes element-wise multiplication. This design preserves in-box observations while suppressing out-of-box regions, thereby reducing background leakage caused by box jitter.

After constructing evidence maps for the past k frames, we concatenate them with the current-frame image along the channel dimension to form the model input

$$\tilde{I}_t = \text{Concat}\,(I_t, E_{t-1}, E_{t-2}, \ldots, E_{t-k}). \tag{3}$$

We then adapt only the input projection layer to accommodate the increased number of channels, while keeping the rest of the backbone unchanged.

This design is driven by two complementary sources of information. The box map M_{t-i} provides a geometric prior from historical predictions, including approximate target location and scale, whereas H_{t-i} preserves the underlying observation evidence. By attenuating out-of-box regions instead of discarding them completely, the model is encouraged to focus on historically plausible locations while still retaining sufficient context from historical observations. In this way, HEI provides observation-conditioned spatial guidance and reduces over-reliance on noisy geometric priors, thereby improving robustness to temporal drift.

In addition to the hard binary box map, we also explored a soft evidence construction based on a Gaussian weighting map centered at the predicted target center. Although the soft box map smooths hard boundaries, our ablation study shows that it weakens the localization effect under aerial clutter. We therefore adopt the hard box map as the default choice for stable localization.

2) Target-Aware Band Selection: When fusing predicted boxes with historical observations, an important design choice is which spectral band should be used as evidence. Using all bands across multiple history frames would substantially increase the input dimensionality and computational overhead. Moreover, band utility is highly scene-dependent: some bands provide strong target–background separability, whereas others are dominated by clutter, shadows, or sensor noise and may amplify errors when historical boxes are imperfect. We therefore select a single sequence-specific band that is both discriminative and background-stable.

For each band l, we consider its reflectance intensity image I_l. On the first frame of each sequence, given the ground-truth box $B = (x, y, w, h)$, where (x, y) denotes the top-left coordinate, and (w, h) denote width and height, we define the foreground region Ω_{fg} and the background region Ω_{bg}. The foreground region is obtained by shrinking the ground-truth box around its center to reduce background contamination, while the local background is defined as a ring region formed by expanding the box with fixed ratios. The shrinkage and expansion ratios are fixed and reported in the Appendix.

For each band l, we compute a Fisher separability score between the target and the local background

$$F(l) = \frac{\left(\mu_{fg}(l) - \mu_{bg}(l)\right)^2}{\sigma^2_{fg}(l) + \sigma^2_{bg}(l) + \epsilon} \tag{4}$$

where $\mu_{fg}(l)$ and $\mu_{bg}(l)$ are the mean intensities of the foreground and background, respectively, and $\sigma^2_{fg}(l)$ and $\sigma^2_{bg}(l)$ denote the corresponding variances. This term favors bands with large interclass differences and small intraclass variability.

If the background is highly variable in a given band, even slight background leakage can substantially degrade the usefulness of historical evidence. We therefore introduce an additional background-stability term

$$S(l) = \frac{1}{\sigma^2_{bg}(l) + \epsilon}. \tag{5}$$

The final band-selection score is defined as

$$\text{score}\,(l) = F(l) \cdot S(l)^{\beta} \tag{6}$$

where $\beta > 0$ controls the emphasis on background stability ($\beta = 1$ by default). Finally, we select the band with the highest score as the historical observation evidence

$$l^* = \arg\max_{l} \text{score}\,(l). \tag{7}$$

This strategy is lightweight, target-adaptive, and helps reduce contamination from background leakage inside predicted boxes.

3) History Length and Temporal Order: In history-guided tracking, the history window length k determines both the temporal order of motion patterns that can be captured and the noise level of the resulting prior. When $k = 1$, the model mainly exploits a first-order state prior from the previous prediction, including target position, size, and aspect ratio. When $k = 2$, it can further access short-term increments such as $\Delta x_t = x_{t-1} - x_{t-2}$, which act as velocity-like cues and support a simple extrapolated prior (e.g., $x_t \approx x_{t-1} + \Delta x_t$) that can then be refined using the current search image. Increasing k further, in principle, enables the tracker to encode higher-order temporal patterns such as acceleration-like changes.

However, a larger k also increases the risk of distribution shift and error accumulation. Since future information is unavailable during inference, the model must be trained under a strictly causal history. Moreover, prediction noise compounds over longer temporal windows, making the constructed prior more vulnerable to drift. We therefore use two historical frames by default, which provides a compact short-term temporal window that captures recent motion trends while limiting stale evidence and error accumulation.

4) Training Strategy With Perturbation and Annealing: Because our historical evidence is constructed from bounding boxes, using ground-truth (GT) boxes throughout training would cause the model to over-rely on overly clean history cues and implicitly assume near-perfect localization. At inference time, however, historical evidence is built from tracker predictions and is inevitably noisy. Similar training–inference

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consistency issues have also been noted in online video tasks such as video instance segmentation [64], where historical representations used during inference may deviate from clean training signals. This discrepancy in history quality introduces a train–test distribution shift, biases the learned temporal prior, and may exacerbate drift over time.

To reduce this gap, we perturb GT boxes during training to mimic inference-time imperfections, and gradually decay the perturbation strength with an annealing schedule to balance robustness and accuracy. Following the jittering strategy used for search-region generation in UNTrack [30] and ODTrack [24], we introduce only an epoch-dependent scaling factor and do not add extra tunable hyperparameters. Given a GT box at epoch e and total epochs E, we define

$$\gamma(e) = \exp\left(-\frac{e}{E/2}\right). \tag{8}$$

The factor $\gamma(e)$ scales the center and size jitter magnitudes: early in training ($e \to 0$), stronger perturbation discourages over-reliance on clean history, while later in training the perturbation decays exponentially to enable finer localization refinement. Because the exponential schedule remains nonzero over the finite training horizon, the perturbation does not vanish completely in practice. This weak residual perturbation reflects the inevitable noise present at inference and helps improve train–test consistency.

We also address a second mismatch at the beginning of a sequence, where the first k frames do not have sufficient history. During training, whenever a required historical index is invalid, we replicate the earliest available evidence, i.e., the closest valid past frame, to fill the missing entries. This exposes the model to missing-history cases without introducing any future information. Overall, the annealed box jitter and start-of-sequence simulation reduce mismatches in both the quality and availability of historical evidence, thereby promoting more reliable and error-aware fusion of temporal cues.

B. TAMTP Module

While HEI improves temporal robustness through lightweight short-term evidence, aerial multispectral tracking also requires stable target identity cues to mitigate accumulated drift and distractor attraction. For tiny targets, the predicted box may contain a substantial amount of surrounding background, making appearance-driven matching vulnerable to nearby distractors. To address this issue, we introduce TAMTP, which maintains anchor-guided target identity cues by combining target–background contrast with compact structural and spectral signatures. These cues are injected as controlled low-dimensional residual updates in the prompt space, and multiple templates are fused with reliability-aware weights to reduce the influence of misaligned or contaminated auxiliary templates during inference.

Formally, given a set of templates $\{(I_i, M_i, \mathcal{T}_i)\}_{i=1}^{K}$, where I_i is the raw template image, M_i is the box-derived mask, and $\mathcal{T}_i$ denotes the patch-embedded template tokens, we first construct a candidate prompt vector $p_i \in \mathbb{R}^C$ for each template. We then use the first template as an anchor and fuse multiple candidate prompts into a single prompt p, while assigning lower weights to auxiliary templates that may be unreliable due to noisy box localization.

1) Target Appearance Prototype: For each template, we construct a target-local background region to suppress irrelevant far-field statistics, since the most confusing distractors usually appear near the target. Specifically, we dilate the box-derived mask to obtain a local-context mask and define the background as the ring region around the target, falling back to the global background when the ring region becomes nearly empty. We then compute an appearance prototype as the difference between masked-pooled target features and masked-pooled local-background features, encouraging the prompt to encode target identity while reducing background leakage.

Given the box mask M_i, we build a local-context mask M_i^{ctx} by dilating M_i on the feature-aligned grid, and define

$$M_i^{\text{bg}} = M_i^{\text{ctx}} - M_i \tag{9}$$

so that the background covers only the neighborhood of the target while excluding the target interior. In extreme cases, such as tiny targets or boundary truncation, where M_i^{bg} becomes nearly empty, we use a fallback definition $M_i^{\text{bg}} = 1 - M_i$.

Based on this construction, we define a target-local appearance contrast as

$$f_i^{\text{app}} = \text{Pool}\left(\mathcal{T}_i, \tilde{M}_i\right) - \text{Pool}\left(\mathcal{T}_i, \tilde{M}_i^{\text{bg}}\right) \tag{10}$$

where $\tilde{M}_i$ is obtained by downsampling the pixel mask M_i to match the spatial resolution of $\mathcal{T}_i$ using area-preserving average pooling, and $\tilde{M}_i^{\text{bg}}$ follows the same construction rule. The masked pooling operation is defined as

$$\text{Pool}\left(\mathcal{T}, \tilde{M}\right) = \frac{\sum_{x,y} \tilde{M}_{x,y} \mathcal{T}_{:,x,y}}{\sum_{x,y} \tilde{M}_{x,y} + \epsilon} \tag{11}$$

where ϵ is a small constant for numerical stability. We then apply layer normalization

$$a_i = \text{LN}\left(f_i^{\text{app}}\right) \tag{12}$$

which reduces magnitude variation across scenes and provides a consistent scale for prompt fusion and template consistency estimation. The vector a_i serves as the primary prompt component, while structural and spectral cues are injected later as residual cues.

2) Target Spatial Structure Embedding: Appearance cues can become unreliable under motion blur or weak texture, where high-contrast background patterns may dominate feature matching. We therefore extract a compact structural signature using a 2-D Haar decomposition and aggregate the high-frequency sub-bands to highlight salient local structure. We further append an ROI mask as an explicit prior and use a lightweight encoder followed by target-minus-background pooling to obtain a target-specific structure embedding.

For each template image I_i, we apply a 2-D Haar decomposition independently to each spectral band and aggregate the high-frequency sub-bands to form a structural response map

$$E_i = |LH(I_i)| + |HL(I_i)| + |HH(I_i)|. \tag{13}$$

We use magnitude responses to reduce sensitivity to coefficient sign changes caused by small shifts, blur, or exposure

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variations, while retaining all three detail sub-bands before aggregation to preserve complementary directional information. The resulting map emphasizes structural intensity and provides a more stable cue when raw texture is unreliable.

We then use a shallow convolutional encoder Φ_e (details are provided in the Appendix) and append an aligned ROI mask as an additional channel

$$F_i^e = \Phi_e\left(\left[E_i; M_i'\right]\right) \tag{14}$$

where M_i' denotes the box-derived mask aligned to the resolution of E_i. Following the same target-centric local-background construction as above, we also obtain the corresponding background mask $M_i'^{,\mathrm{bg}}$. Finally, we compute a target-specific structure embedding using masked average pooling with foreground–background differencing

$$e_i = \mathrm{Pool}\left(F_i^e, M_i'\right) - \mathrm{Pool}\left(F_i^e, M_i'^{,\mathrm{bg}}\right). \tag{15}$$

3) Target Spectral Signature Embedding: To capture material-related cross-band responses that are more stable than raw texture, we construct a spectral signature by normalizing per-pixel magnitudes to reduce illumination effects and performing a 1-D Haar decomposition along the wavelength axis. Instead of applying early ROI pooling, which may wash out local contamination, we retain the spectral decomposition at each spatial location and use a lightweight local encoder with the ROI mask appended. A target-minus-background pooling then yields a compact target spectral embedding.

Concretely, we construct a **spectral signature** feature that emphasizes cross-band relations while suppressing global magnitude variation. We first apply per-pixel scale normalization to reduce illumination effects, and then perform a 1-D Haar decomposition along the wavelength axis to obtain a low-frequency trend and high-frequency details, which are concatenated into a spectral-shape map

$$(L_i, D_i) = \mathrm{Haar1D}\left(I_i\right), \quad S_i = [L_i; D_i]. \tag{16}$$

We keep this spectral decomposition at each spatial location, allowing subsequent local modeling to handle occlusion or boundary contamination instead of discarding such information through early ROI pooling. Following the structure branch, we align S_i to the same spatial resolution and use a lightweight local encoder Φ_s (sharing the same architecture design as Φ_e), with the ROI mask appended as an explicit prior

$$F_i^s = \Phi_s\left(\left[S_i'; M_i'\right]\right). \tag{17}$$

Using the same target-centric local-background construction as in Section III-B2, we extract the target spectral embedding via masked average pooling with local-background subtraction

$$h_i = \mathrm{Pool}\left(F_i^s, M_i'\right) - \mathrm{Pool}\left(F_i^s, M_i'^{,\mathrm{bg}}\right). \tag{18}$$

4) Multitemplate Prompt Fusion: We treat the appearance prototype a_i as the primary identity representation, while using the structural cue e_i and the spectral cue h_i as complementary evidence. Specifically, we project e_i and h_i to the prompt channel dimension and normalize them

$$u_i^e = \mathrm{LN}\left(W_e e_i\right), \quad u_i^s = \mathrm{LN}\left(W_s h_i\right) \tag{19}$$

where $W_e \in \mathbb{R}^{C\times d_e}$ and $W_s \in \mathbb{R}^{C\times d_s}$, with $d_e, d_s \ll C$. This design restricts the injected updates to low-dimensional subspaces, reducing direct competition between the structural/spectral branches and the appearance backbone.

Because the reliability of structural and spectral evidence varies across scenes, we adopt a lightweight gated fusion to control the magnitudes of the two residuals, yielding a candidate prompt for each template

$$p_i = a_i + \alpha_e u_i^e + \alpha_s u_i^s \tag{20}$$

where α_e and α_s are learnable scalar gates. This parameterization allows the model to behave like a pure appearance prompt at the beginning of training and gradually learn when, and to what extent, structural and spectral residual cues should be introduced.

In the multitemplate setting, only the first-frame box is guaranteed to be accurate. We therefore use the first template as a long-term identity anchor and compute the cosine consistency of each auxiliary template with respect to this anchor. The three consistencies are averaged to produce a template reliability score

$$s_i = \frac{1}{3}\left(\cos\left(a_i, a_1\right) + \cos\left(e_i, e_1\right) + \cos\left(h_i, h_1\right)\right). \tag{21}$$

We then convert these scores into normalized reliability weights via a softmax, while adding a fixed positive bias to the anchor template to ensure a robust fallback under uncertainty

$$r_i = \mathrm{softmax}_i\left(s_i + \delta\,\mathbb{I}[i=1]\right), \quad \delta > 0 \tag{22}$$

where $\mathbb{I}[\cdot]$ denotes the indicator function. Finally, the multiple candidate prompts are fused into a single prompt token

$$p = \sum_{i=1}^{K} r_i\, p_i. \tag{23}$$

The resulting prompt is injected into the Transformer blocks as a target-aware conditioning signal. This fusion strategy exploits auxiliary templates when they are reliable and falls back to the anchor when they are noisy, without introducing additional thresholds or temperature terms. To better match inference conditions, we also jitter the auxiliary template boxes used for prompt extraction during training to simulate template misalignment.

C. Training Objective

Unless otherwise stated, the target-search interaction, prediction head, and training objective follow UNTrack [30]. Our modifications are limited to the construction of historical evidence, the generation and fusion of target-aware prompts, and their integration into the original tracking pipeline.

Accordingly, we do not introduce any new loss term. HEATTrack is trained using the same optimization objective as UNTrack, and the proposed HEI and TAMTP modules are learned jointly through the original tracking supervision. This design ensures that the performance gains come from improved temporal evidence modeling and target-aware spectral prompting, rather than from additional objective engineering.

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IV. Experiments and Analysis

A. Datasets

We evaluate HEATTrack on two public multispectral SOT benchmarks. MUST serves as our primary benchmark because it is specifically designed for aerial multispectral tracking under moving-platform disturbances, while MSITrack is used as a complementary benchmark for broader multispectral tracking scenarios.

1) MUST Dataset: We mainly evaluate on MUST [30], a public benchmark for aerial multispectral SOT. MUST contains 250 video sequences with approximately 43K frames. Each frame provides eight spectral bands spanning 395–950 nm, with a spatial resolution of 1280 $\times$ 960 pixels. The dataset is designed to balance spectral richness and temporal continuity, and it includes 12 challenging attributes. We follow the official split, using 160 sequences for training and 90 for testing.

2) MSITrack Dataset: As a complementary evaluation beyond the primary aerial benchmark, we also evaluate on MSITrack [31], a public multispectral SOT benchmark in broader scenes. MSITrack contains 300 video sequences with approximately 129K frames, covering 55 target categories across diverse scenes and time periods. Similar to MUST, it provides eight spectral bands spanning 395–950 nm. We follow its official split, using 220 sequences for training and 80 for testing.

B. Evaluation Metrics

We evaluate tracking performance from two complementary perspectives: region overlap quality (**SR** and **AUC**) and center localization accuracy (**Pre** and **PreN**). Consider a sequence of length T, where the predicted box and ground-truth box at frame t are denoted by $\mathbf{B}_t$ and $\mathbf{G}_t$, respectively. Their center coordinates are denoted by $\mathbf{c}(\mathbf{B}_t)$ and $\mathbf{c}(\mathbf{G}_t)$. All metrics are first computed at the sequence level and then averaged over all sequences to obtain the final benchmark score.

1) Success Rate: Success rate (SR) measures overlap quality based on the intersection-over-union (IoU) between the predicted and ground-truth boxes. Given an IoU threshold $\tau \in [0, 1]$, we define the per-frame success indicator as

$$s_t(\tau) = \mathbb{I}\left(\mathrm{IoU}\left(\mathbf{B}_t, \mathbf{G}_t\right) > \tau\right) \tag{24}$$

where $\mathbb{I}(\cdot)$ is the indicator function. The sequence-level SR is then

$$\mathrm{SR}(\tau) = \frac{1}{T}\sum_{t=1}^{T} s_t(\tau). \tag{25}$$

Following UNTrack [30], we report $\mathrm{SR}_{0.5}$ and $\mathrm{SR}_{0.75}$. The former reflects robustness under moderate overlap requirements, while the latter emphasizes stricter localization quality.

2) Area Under Curve: Since a single-threshold SR only reflects performance at one overlap strictness level, we vary τ over $[0, 1]$ to construct the success plot and compute the AUC as

$$\mathrm{AUC} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{SR}(\tau_k) \tag{26}$$

where $\{\tau_k\}_{k=1}^{K}$ is uniformly sampled thresholds from 0 to 1 at an interval of 0.05. AUC measures the average SR over the full IoU threshold range and therefore provides a more comprehensive assessment of overlap quality.

3) Precision (Pre): Precision evaluates localization accuracy by measuring the Euclidean distance between the predicted and ground-truth box centers. The center location error at frame t is defined as

$$d_t = \left\|\mathbf{c}\left(\mathbf{B}_t\right) - \mathbf{c}\left(\mathbf{G}_t\right)\right\|_2. \tag{27}$$

Given a distance threshold δ, the precision is

$$\mathrm{Pre}(\delta) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}(d_t < \delta). \tag{28}$$

Following UNTrack [30], we report Pre_{20}, i.e., precision at $\delta = 20$ pixels.

4) Normalized Precision (PreN): Fixed-pixel precision is sensitive to target scale and image resolution. To obtain a scale-invariant localization measure, we normalize the center coordinates by the ground-truth box size. Specifically, let

$$\tilde{\mathbf{c}}\left(\mathbf{B}_t\right) = \mathbf{c}\left(\mathbf{B}_t\right) \oslash \left(w_t^G, h_t^G\right) \tag{29}$$

$$\tilde{\mathbf{c}}\left(\mathbf{G}_t\right) = \mathbf{c}\left(\mathbf{G}_t\right) \oslash \left(w_t^G, h_t^G\right) \tag{30}$$

where (w_t^G, h_t^G) denotes the width and height of the ground-truth box $\mathbf{G}_t$, and $\oslash$ denotes element-wise division. The normalized center error is then

$$\tilde{d}_t = \left\|\tilde{\mathbf{c}}\left(\mathbf{B}_t\right) - \tilde{\mathbf{c}}\left(\mathbf{G}_t\right)\right\|_2. \tag{31}$$

Given a normalized threshold γ, the normalized precision is

$$\mathrm{PreN}(\gamma) = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}\left(\tilde{d}_t \le \gamma\right). \tag{32}$$

Following UNTrack [30], we use the PreN (normalized precision at $\gamma = 0.20$) to evaluate the trackers.

C. Implementation Details

We build HEATTrack on top of UNTrack [30] and keep the core Transformer blocks, prediction head, and training objective unchanged. We use ViT-B/16 (patch size 16) as the backbone and optimize the model with AdamW. The ViT backbone is fully fine-tuned during training. The search region is resized to 384 $\times$ 384, and the template size is set to 192 $\times$ 192. We train the model for 50 epochs with a per-GPU batch size of 4 on 4 NVIDIA RTX 4090 GPUs, resulting in an effective batch size of 16. The initial learning rate is set to 4×10^{-4} and decayed by a factor of 10 after epoch 30. Unless otherwise stated, we fix the random seeds in the reproduced training runs and use the same training and evaluation protocol for the reproduced baselines and HEATTrack variants.

1) Input Protocol: UNTrack takes two search frames (current and previous) as input, whereas our method uses a single current-frame search image augmented with HEI evidence maps. For the main comparison, UNTrack and our reproduced UNTrack* follow the original UNTrack input protocol. UNTrack* follows the original UNTrack configuration as closely as possible, with the main adjustment being the initial

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TABLE I

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE MUST TEST SET. **UNTrack*** DENOTES OUR REPRODUCED UNTRACK BASELINE UNDER OUR TRAINING ENVIRONMENT. **Ours**† IS TRAINED WITH LEARNING RATE 1×10^{-4}, WHILE **Ours** IS TRAINED WITH LEARNING RATE 4×10^{-4}. THE REMAINING RESULTS ARE TAKEN FROM THE ORIGINAL UNTRACK PAPER. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	AUC ↑	$SR_{0.5}$ ↑	$SR_{0.75}$ ↑	Pre ↑	PreN ↑
SiamRPN [1]	38.9	49.5	22.3	56.7	50.1
SiamMask [2]	39.1	51.0	25.3	56.1	52.8
DiMP [3]	47.3	60.7	31.6	69.8	57.1
OSTrack [4]	55.1	69.6	44.1	73.3	68.8
ODTrack [24]	52.7	67.3	38.8	71.3	66.2
ZoomTrack [27]	53.3	67.7	40.4	72.3	66.7
UNTrack [30]	59.7	75.8	46.2	79.2	74.8
UNTrack* [30]	62.8	78.6	53.3	82.5	77.5
Ours†	<u>66.0</u>	<u>82.7</u>	<u>58.5</u>	<u>85.4</u>	<u>82.3</u>
Ours	**67.1**	**83.7**	**59.5**	**86.9**	**83.6**

learning rate for stable reproduction under our environment. We additionally report a Search = 1 matched control in the ablation study to separate the effect of input protocol from the proposed evidence construction. For transparency, we provide the detailed reproduction configuration of UNTrack* in the Appendix, including the input protocol, image sizes, optimizer, learning rate, and training split.

We use only the ViT backbone initialization without additional pretrained models or external data. Following UNTrack [30], we adopt its parameter reconstruction strategy for multispectral inputs and modify only the input projection layer to accept the concatenated HEI channels, initializing the added channels by replicating the original patch embedding weights.

Unless otherwise stated, the target-search interaction, prediction head, and optimization objective follow UNTrack [30]; no additional loss term is introduced. The proposed HEI and TAMTP modules are learned jointly under the original tracking supervision. Additional implementation details are provided in the Appendix.

D. Main Results

1) Main Results on MUST Dataset: On MUST, our primary aerial multispectral benchmark, Table I shows that HEATTrack achieves the best overall performance across both overlap-based and localization-based metrics. Compared with the reproduced UNTrack* baseline under our training setting, HEATTrack improves AUC from 62.8 to 67.1 (+4.3) and PreN from 77.5 to 83.6 (+6.1), indicating not only stronger overlap quality but also better scale-invariant localization accuracy under aerial drift and scale variation. The attribute-wise results in Table II further show that the gains are concentrated on drift-related challenges, including camera motion, fast motion, motion blur, and partial occlusion, which is consistent with the design of explicit history evidence and target-aware identity conditioning.

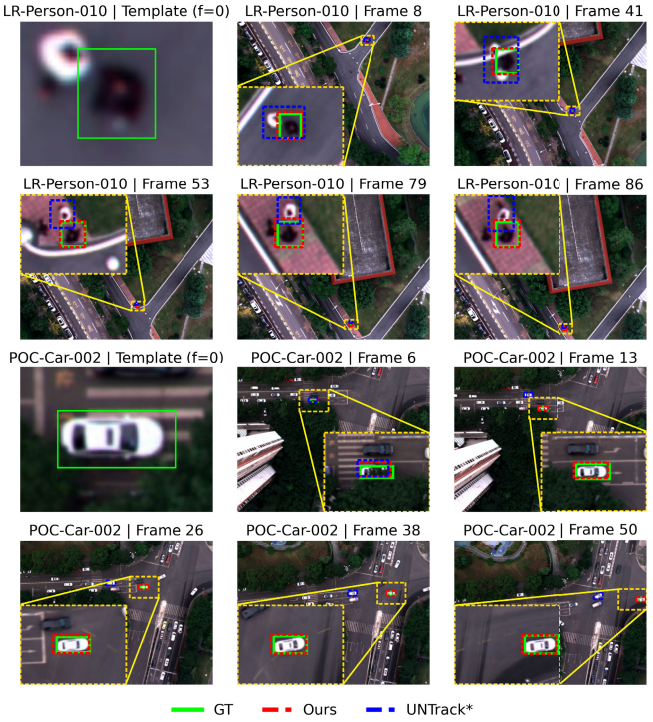


Fig. 2. Qualitative comparison of tracking results on representative MUST test sequences. Our method produces more stable predictions in the presence of similar objects, whereas UNTrack suffers from tracking failure.

We observe clear improvements under appearance-confusing conditions, including BC (57.7 → 58.1), SC (66.5 → 69.6), and SOB (59.7 → 64.4). As shown in Fig. 2, our method yields more stable tracking results in the presence of similar objects. This is consistent with the TAMTP design, which strengthens identity discrimination when tiny targets are mixed with surrounding background pixels. Such cases often fail through gradual attraction to nearby distractors, and the tighter localization quality is reflected in the improved strict-overlap and scale-normalized localization metrics reported in Table I.

Fig. 3 provides a representative temporal IoU curve consistent with the attribute-wise results in Table II. On sequence BC-Bicycle-002, both methods experience a temporary failure around frame 8, but our method reacquires the target afterward, whereas UNTrack* remains at nearly zero IoU. This suggests that historical evidence provides useful temporal guidance that helps the tracker recover from transient failures, rather than remaining trapped in the crop-update feedback loop.

2) Main Results on MSITrack Dataset: On MSITrack, which serves as a complementary benchmark beyond the primary aerial setting, our method also shows strong and consistent improvements. As shown in Table III, HEATTrack achieves 56.2 AUC and 68.7 PreN. Compared with our reproduced UNTrack* baseline on MSITrack, HEATTrack improves AUC from 50.5 to 56.2 (+5.7) and PreN from 62.6 to 68.7 (+6.1). These results indicate that the proposed design remains effective in broader multispectral tracking scenarios.

Fig. 4 reports the attribute-wise AUC on MSITrack. Our method achieves the best result on all listed challenges,

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TABLE II

COMPARISON OF AVERAGE AUC (%) UNDER DIFFERENT CHALLENGE ATTRIBUTES ON MUST. BC: BACKGROUND CLUTTER; CM: CAMERA MOTION; FM: FAST MOTION; FOC: FULL OCCLUSION; IV: ILLUMINATION VARIATION; LR: LOW RESOLUTION; MB: MOTION BLUR; OV: OUT-OF-VIEW; POC: PARTIAL OCCLUSION; SV: SCALE VARIATION; SC: SIMILAR COLOR; AND SOB: SIMILAR OBJECT. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	BC	CM	FM	FOC	IV	LR	MB	OV	POC	SV	SC	SOB
UNTrack	51.7	59.1	56.8	54.9	64.3	52.7	62.3	42.3	59.6	53.5	61.8	54.7
UNTrack*	57.7	63.4	63.8	56.4	**70.5**	54.9	65.0	44.7	62.4	54.1	66.5	59.7
Ours	**58.1**	**71.3**	**68.9**	**60.1**	69.8	**57.0**	**77.4**	**51.6**	**66.3**	**56.8**	**69.6**	**64.4**

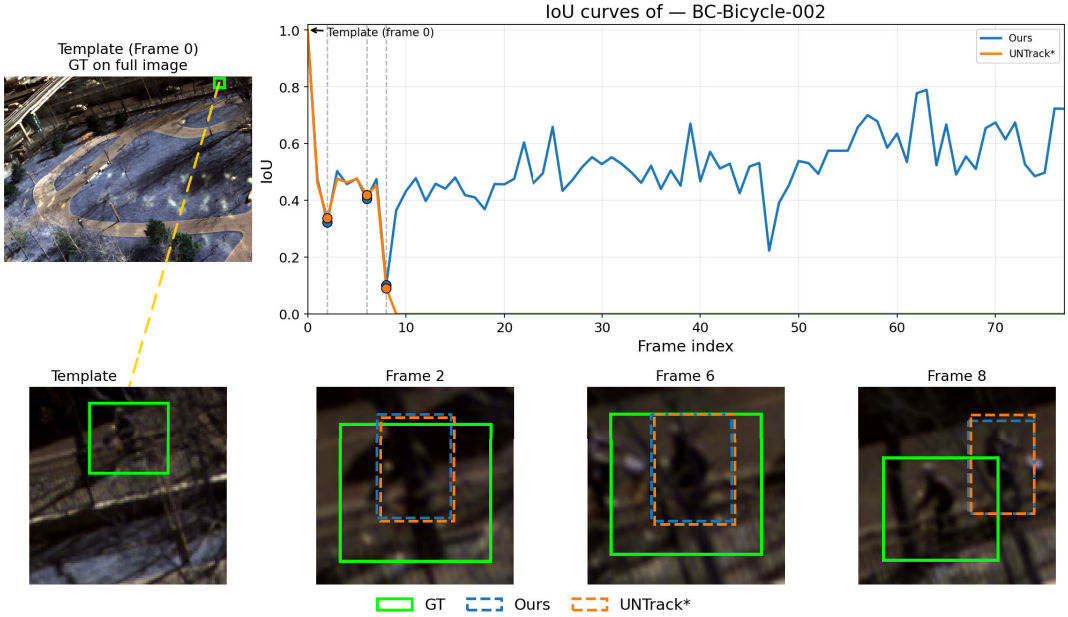


Fig. 3. Comparison of IoU curves on MUST (sequence BC-Bicycle-002). UNTrack* loses the target after frame 8 and remains at nearly zero IoU, whereas our method reacquires the target in subsequent frames.

TABLE III

EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE MSITRACK TEST SET. UNTRACK* DENOTES OUR REPRODUCED UNTRACK BASELINE UNDER THE SAME TRAINING ENVIRONMENT. THE REMAINING NON-HEATTRACK RESULTS ARE TAKEN FROM MSITRACK [31]. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	AUC ↑	$SR_{0.5}$ ↑	$SR_{0.75}$ ↑	Pre ↑	PreN ↑
OSTrack-256 [4]	42.0	50.9	34.2	55.7	50.5
OSTrack-384 [4]	43.1	52.5	36.1	56.2	52.4
LMTrack [29]	43.2	52.4	39.7	55.2	52.5
ZoomTrack [27]	44.3	54.7	35.1	59.0	54.4
NeighborTrack [28]	44.8	55.2	35.7	59.7	54.6
EVPTrack [25]	46.2	55.8	43.9	58.3	56.0
AQATrack [26]	47.1	56.7	44.7	58.5	56.9
UNTrack [30]	47.3	58.4	40.5	61.8	57.8
GRM [65]	47.8	58.1	40.8	63.0	57.8
UNTrack* [30] (ours)	50.5	62.3	44.9	64.1	62.6
Ours	**56.2**	**69.1**	**51.4**	**70.9**	**68.7**

TABLE IV

ABLATION ON THE TWO MAIN MODULES, HEI AND TAMTP, ON THE MUST TEST SET

HEI	**TAMTP**	**AUC**	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
-	-	62.75	78.58	53.32	82.46	77.50
✓	-	66.56	83.49	58.55	86.07	**83.82**
-	✓	65.77	82.30	57.69	85.01	83.24
✓	✓	**67.05**	**83.71**	**59.50**	**86.88**	83.55

indicating robust performance across diverse multispectral scenarios. The largest gains appear on SV (62.2 versus 50.3) and POC (52.4 versus 43.4), suggesting stronger robustness under significant scale variation and partial occlusion, where matching ambiguity and distractor interference are especially severe. We also observe clear improvements on LR (54.8 versus 48.5), showing better tolerance to tiny low-resolution targets with weak texture cues. Moreover, the consistent gains on SC (48.1 versus 46.9) and SOB (43.5 versus 39.0) reflect improved resistance to similar-color and similar-object distractors, which are common failure modes in MSITrack. Overall, these attribute-wise results support that HEATTrack provides reliable gains on both the primary aerial benchmark and the complementary distractor-heavy MSITrack benchmark.

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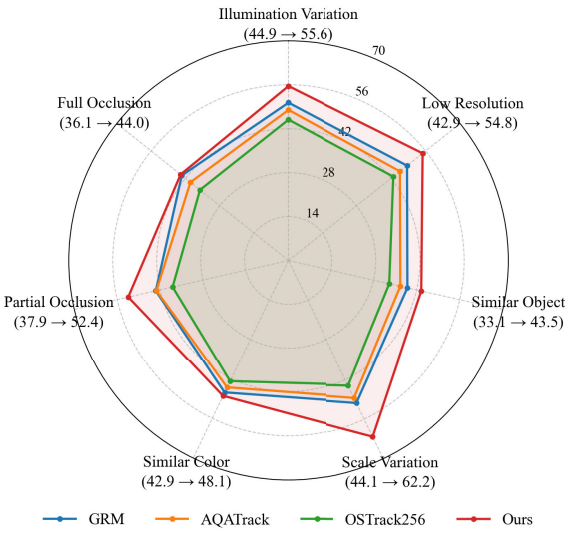


Fig. 4. Attribute-wise AUC comparison of different methods on the MSITrack test set.



Fig. 5. Classification score maps on search-area crops. Warmer colors indicate higher target confidence. We overlay the predicted box (red) and ground truth (cyan) in the same crop coordinates. The left column corresponds to UNTrack*, while the right column shows UNTrack* augmented with TAMTP under the same setting.

E. Ablation Studies and Further Analysis

We conduct ablation studies on MUST using UNTrack* as the reference baseline. Our goal is to separately validate: 1) whether explicit history evidence improves temporal robustness beyond implicit history usage and 2) whether target-aware prompting strengthens identity discrimination under distractors. We therefore report both overlap-based metrics (AUC/SR) and localization-based metrics (Pre/PreN), and analyze where each module contributes most.

When evaluating the internal components of HEI and TAMTP separately, we disable the other module accordingly. Specifically, TAMTP is removed when studying HEI-related factors such as band selection and the number of historical frames, while HEI is disabled when evaluating the three components of TAMTP.

1) Contributions of the Two Main Modules: We first examine the independent contributions of **HEI** and **TAMTP** in Table IV, as well as their complementarity. Using UNTrack* as the baseline, introducing history evidence alone yields substantial gains across all metrics (+3.81 AUC, +3.61 Pre, and +6.32 PreN). These improvements confirm that explicit history evidence provides effective target-state priors, including location, scale, and short-term motion trends, which help suppress temporal drift. In particular, the large gains on the stricter overlap metric $SR_{0.75}$ and on PreN suggest that the benefit goes beyond coarse recovery and improves high-quality overlap together with scale-normalized localization accuracy.

Using TAMTP alone also brings consistent improvements (+3.02 AUC, +2.55 Pre, and +5.74 PreN), although with a smaller overall margin than HEI. Its gains are particularly clear on the localization-based metrics Pre and PreN, suggesting that this module strengthens identity discrimination through multitemplate and multispectral target-specific cues, thereby mitigating background attraction and false matches in cluttered scenes.

When both modules are enabled, the model achieves the best overall performance. Compared with using HEI alone, the joint setting still brings additional gains in AUC (+0.49), $SR_{0.75}$ (+0.95), and Pre (+0.81), while maintaining similarly strong PreN performance (83.55 versus 83.82). This result supports the intended division of roles between the two modules: HEI mainly provides short-term temporal evidence for stabilizing recent localization, whereas TAMTP contributes anchor-guided identity cues for suppressing distractors and maintaining target discrimination over time.

To further illustrate the effect of the target-aware multitemplate prompt under cluttered backgrounds, we visualize the classifier score maps on search-area crops in Fig. 5. We observe that UNTrack* is more prone to background attraction over time: the response peak gradually becomes broader, more diffuse, or shifts away from the target, and is more easily distracted by nearby high-contrast structures, shadow textures, or similar instances. In Fig. 5, the peak progressively drifts toward a distractor vehicle, leading to systematic localization bias; once such an error enters the crop-update feedback loop, it tends to be reinforced and becomes difficult to recover.

In contrast, the model equipped with TAMTP maintains sharper and more concentrated responses that remain within the target region across frames. Overall, these results further support the complementarity of the two modules: history evidence improves temporal robustness, while target-aware prompting strengthens identity discrimination in cluttered backgrounds, and together they yield more accurate and robust aerial multispectral tracking.

2) Matched Input-Protocol Control: To separate the effect of temporal input protocol from the proposed evidence construction, we additionally compare UNTrack* under Search = 1 and Search = 2 on MUST. As shown in Table V, the two UNTrack* variants obtain similar AUC scores (62.33 versus 62.75), indicating that the gain of HEI is not

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TABLE V

MATCHED INPUT-PROTOCOL CONTROL ON MUST. SEARCH = 2 DENOTES THE ORIGINAL UNTRACK PROTOCOL USING CURRENT AND PREVIOUS SEARCH FRAMES, WHILE SEARCH = 1 USES ONLY THE CURRENT SEARCH FRAME

Method	Search	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
UNTrack*	2	62.75	78.58	53.32	82.46	77.50
UNTrack*	1	62.33	78.21	51.80	81.98	77.78
Ours (Only HEI)	1	66.56	83.49	58.55	86.07	83.82
Ours	1	**67.05**	**83.71**	**59.50**	**86.88**	83.55

TABLE VI

EFFICIENCY COMPARISON OF UNTRACK AND HEATTRACK VARIANTS

Method	Search	GFLOPs ↓	FPS ↑	Params (M) ↓
UNTrack*	1	84.0	69.5	113.28
Ours (Only HEI)	1	84.4	69.3	113.67
Ours (Only TAMTP)	1	85.1	50.5	113.52
Ours	1	85.5	50.3	113.92
UNTrack*	2	129.7	48.9	113.72

TABLE VII

ABLATION STUDY ON THE THREE COMPONENTS OF TAMTP: APPEARANCE PROTOTYPE, SPATIAL STRUCTURE, AND SPECTRAL SIGNATURE. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	**AUC**	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
UNTrack*	62.75	78.58	53.32	82.46	77.50
+ Appearance	64.78	80.82	<u>57.44</u>	84.12	80.52
+ Appearance + Spatial	<u>65.38</u>	81.64	57.29	<u>84.96</u>	<u>81.36</u>
+ Appearance + Spectral	65.29	<u>81.80</u>	57.17	84.34	81.22
Full TAMTP	**65.77**	**82.30**	**57.69**	**85.01**	**83.24**

TABLE VIII

ABLATION OF TEMPLATE FUSION STRATEGIES IN TAMTP ON MUST. ALL VARIANTS USE THE FULL APPEARANCE, SPATIAL, AND SPECTRAL PROMPT CUES; ONLY THE TEMPLATE FUSION STRATEGY IS CHANGED. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Fusion strategy	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
Anchor-only	62.80	78.63	54.71	81.55	78.75
Equal-weight	63.77	<u>80.24</u>	54.55	83.12	79.92
Reliability w/o bias	<u>64.30</u>	80.18	<u>55.87</u>	<u>83.74</u>	<u>80.01</u>
Full TAMTP	**65.77**	**82.30**	**57.69**	**85.01**	**83.24**

explained by simply changing the number of search frames. Replacing the raw previous-search input with explicit evidence maps improves AUC to 66.56 and PreN to 83.82, and the full HEATTrack further improves AUC to 67.05. Therefore, the improvement is mainly attributable to how temporal information is represented and filtered, rather than to simply changing the number of search frames.

3) Complexity and Speed Analysis: To quantify computational overhead, we report parameter count, GFLOPs, and FPS under the same runtime environment in Table VI. Since wall-clock FPS depends on the runtime environment, we report tracker FPS only as a relative reference, while parameter count and GFLOPs serve as the primary hardware-independent measures. Compared with the matched UNTrack Search = 1 control, HEI introduces negligible overhead (84.0 → 84.4 GFLOPs, 113.28 → 113.67-M parameters, and 69.5 → 69.3 FPS), showing that short-term evidence construction is lightweight. TAMTP accounts for most of the runtime overhead, although its parameter and GFLOP increases remain small. Compared with the original UNTrack Search = 2 protocol, the full model still requires fewer GFLOPs (85.5 versus 129.7) and achieves slightly higher FPS (50.3 versus 48.9). This reflects the division of roles in HEATTrack: HEI provides lightweight short-term evidence, whereas TAMTP contributes the main identity-prompting capacity and runtime cost.

4) Effectiveness of the Three Components in TAMTP: We ablate the three components of the proposed TAMTP: 1) the appearance prototype; 2) the spatial structure signature; and 3) the spectral signature. All variants are built on top of UNTrack* under the same training and evaluation protocol on MUST. As shown in Table VII, introducing the appearance prototype alone already brings clear improvement over UNTrack* (+2.03 AUC and +3.02 PreN), indicating that explicitly encoding target-versus-local-background contrast in the prompt space is effective for reducing drift and improving localization quality.

Adding either auxiliary cue on top of the appearance prototype further improves performance. The spatial structure branch yields slightly larger gains in AUC and Pre than the spectral branch, suggesting that high-frequency structural evidence is particularly helpful in aerial scenarios with motion blur and strong background texture. In contrast, the spectral branch provides comparable gains on overlap-based metrics, indicating that band-wise reflectance patterns offer complementary identity information beyond appearance cues alone.

Combining all three components achieves the best overall performance. Full TAMTP attains the highest AUC and SRs, and also provides a notable gain in PreN, improving it from 77.50 (UNTrack*) to 83.24 (+5.74). This indicates that TAMTP does not merely increase prompt capacity; instead, it combines appearance, structure, and spectral signatures as complementary identity cues, which is especially useful when tiny targets are partially contaminated by local background.

5) Template Fusion Strategy in TAMTP: While the above ablation verifies the complementary roles of appearance, spatial, and spectral cues, TAMTP also relies on how multiple template prompts are fused over time. Auxiliary templates can provide useful temporal updates, but they may also be contaminated by tracking noise, box misalignment, or nearby distractors. We therefore isolate the template fusion strategy to examine whether auxiliary prompts should be discarded, uniformly averaged, or adaptively weighted according to their reliability. All variants in Table VIII use the full appearance, spatial, and spectral prompt cues, and differ only in how the three template prompts are fused. The anchor-only variant uses only the first-frame template prompt, the equal-weight variant averages all template prompts, and the reliability-aware

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TABLE IX

ABLATION OF DIFFERENT HISTORY EVIDENCE CONSTRUCTIONS ON MUST. "BOX" DENOTES BOX-ONLY HISTORY, "OBS" DENOTES HISTORICAL OBSERVATION, AND "JIT" DENOTES THE ANNEALED JITTER TRAINING STRATEGY

Method	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
UNTrack*	62.75	78.58	53.32	82.46	77.50
Box	60.07	75.70	49.19	79.94	75.08
Box+jit	62.55	78.22	55.22	81.01	78.15
Box+obs+jit	**66.56**	**83.49**	**58.55**	**86.07**	**83.82**
Soft box+obs+jit	65.24	81.56	57.45	84.08	82.16

TABLE X

ABLATION OF HISTORICAL-BOX PERTURBATION STRATEGIES FOR HEI ON MUST. ALL VARIANTS USE THE SAME BOX + OBSERVATION EVIDENCE; ONLY THE TRAINING PERTURBATION STRATEGY IS CHANGED

Training strategy	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
No jitter	61.37	76.95	53.86	79.49	76.82
Fixed jitter	63.21	78.98	55.94	81.79	79.34
Annealed jitter	**66.56**	**83.49**	**58.55**	**86.07**	**83.82**

TABLE XI

ROBUSTNESS TO CONTROLLED PERTURBATIONS OF HISTORICAL BOXES ON MUST. BOTH VARIANTS ARE TRAINED WITH ANNEALED JITTER. THE PERTURBATION IS APPLIED ONLY TO THE HISTORICAL BOXES USED FOR EVIDENCE CONSTRUCTION DURING INFERENCE

Method	0%	5%	10%	15%	20%
AUC ↑					
Box-only	62.55	62.38	61.79	61.31	60.23
Box+obs	**66.56**	**66.37**	**66.05**	**65.51**	**64.90**
PreN ↑					
Box-only	78.15	78.23	77.42	76.33	75.18
Box+obs	**83.82**	**83.50**	**83.01**	**82.40**	**81.74**

variant computes template weights from cue consistency. The full TAMTP further introduces an anchor bias to preserve the first-frame template as a reliable fallback.

As shown in Table VIII, using only the anchor prompt yields limited improvement, with 62.80 AUC and 78.75 PreN. This indicates that the first-frame prompt alone cannot sufficiently adapt to appearance changes, scale variations, and distractors in aerial tracking. Equal-weight fusion improves over anchor-only prompting, showing that auxiliary templates provide complementary target cues. However, it remains inferior to reliability-aware fusion, suggesting that noisy auxiliary templates should not be fused indiscriminately.

Removing the anchor bias from reliability-aware fusion also leads to clear degradation, especially on PreN (83.24 → 80.01). This degradation suggests that reliability estimation alone is insufficient when all auxiliary templates become uncertain, and the first-frame anchor is still needed as a conservative identity fallback. This result indicates that the first-frame anchor provides an important long-term identity reference when auxiliary templates are contaminated by tracking noise or distractors. The full TAMTP achieves the best performance, validating the combination of anchor-guided identity preservation, auxiliary template adaptation, and reliability-aware fusion.

6) History Evidence Construction: Beyond the main-module ablations, we compare different history evidence constructions in Table IX and Fig. 6. Box-only history degrades performance, indicating a clear train–test mismatch: clean boxes during training make the geometric prior appear overly reliable, while noisy predicted boxes at inference can be over-trusted and accelerate drift. Adding annealed jitter partly alleviates this mismatch, bringing AUC back close to the baseline and improving strict-overlap quality, but the gains remain limited because box-only history still lacks observation evidence. By fusing jittered historical boxes with selected-band observations, Box + obs + jit improves all metrics, showing that short-term temporal evidence is useful only when geometric history is coupled with observation-conditioned target–background evidence. The Gaussian soft map performs worse than the hard box map, likely because soft weighting diffuses already weak target responses for tiny aerial targets, whereas the hard box map preserves sharper localization contrast.

Since Table IX changes both the evidence source and perturbation, Table X isolates the perturbation schedule under the same box + observation evidence. No jitter suffers from a clear train–test mismatch, fixed jitter improves robustness, and annealed jitter achieves the best performance. This suggests that stronger early perturbations improve tolerance to noisy history, while later decay supports more precise localization.

7) Robustness and Visualization of History Evidence: Fig. 7 visualizes the evidence maps used by HEI. Compared with the box-only prior, HEI retains target-related historical observations inside the predicted region while attenuating the surrounding background, explaining the stronger performance of Box + obs + jit over Box + jit. The comparison also shows that the box-only prior carries only geometric support, whereas HEI provides localized appearance/spectral evidence from historical observations; this is why the evidence map acts as meaningful temporal guidance rather than merely adding extra input channels.

We further perturb only the historical boxes used for evidence construction during inference, while keeping the current search image, tracker state, and model weights unchanged. As shown in Table XI, Box + obs consistently maintains higher AUC and PreN under increasing perturbation levels. From 0% to 20% perturbation, Box-only drops by 2.32 AUC and 2.97 PreN, whereas Box + obs drops by 1.66 AUC and 2.08 PreN, indicating that historical observations make short-term temporal cues less sensitive to noisy geometric priors.

8) Band Selection Strategy for Historical Observations: When fusing historical predictions with historical observations, choosing which spectral band(s) to use is nontrivial: more channels do not necessarily help, since the historical evidence is conditioned on imperfect predicted states, and redundant or noisy bands may amplify bias. We compare four options under the same setting: 1) fixed three-band fusion

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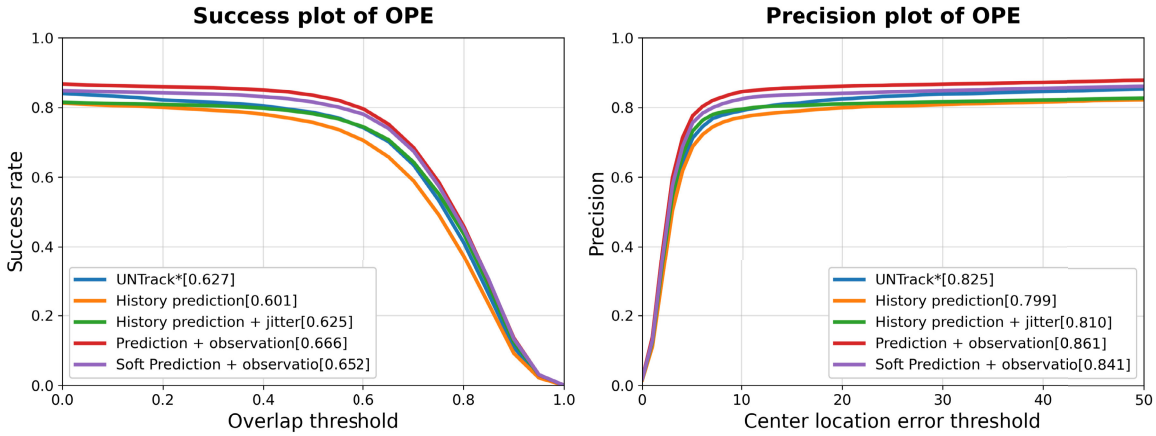


Fig. 6. One-pass evaluation (OPE) curves of variants with different history evidence constructions on MUST.

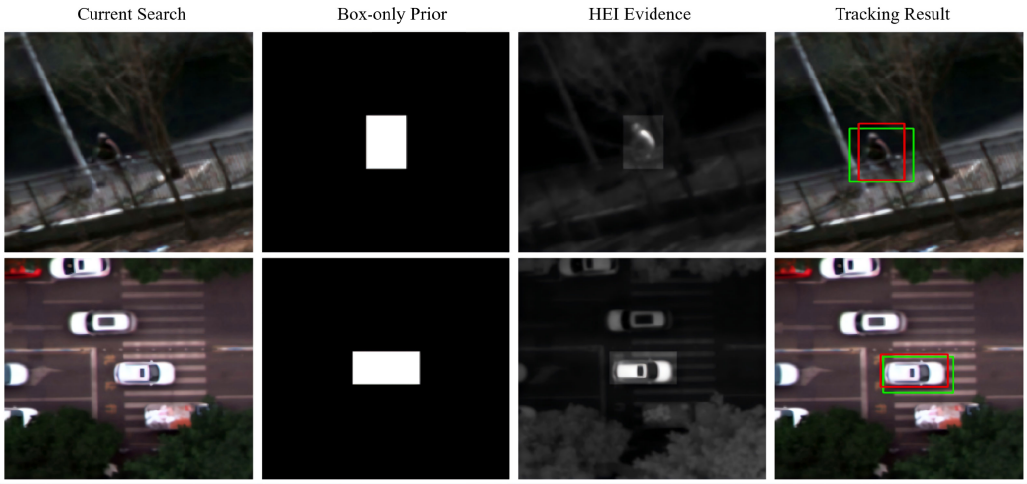


Fig. 7. Visualization of HEI evidence maps on MUST. We show the actual 384 × 384 search-region crops used by the tracker. The box-only prior encodes only the historical geometric box, whereas HEI preserves target-related historical observations inside the predicted region and attenuates irrelevant background. Red and green boxes denote the predicted box and the ground-truth box, respectively.

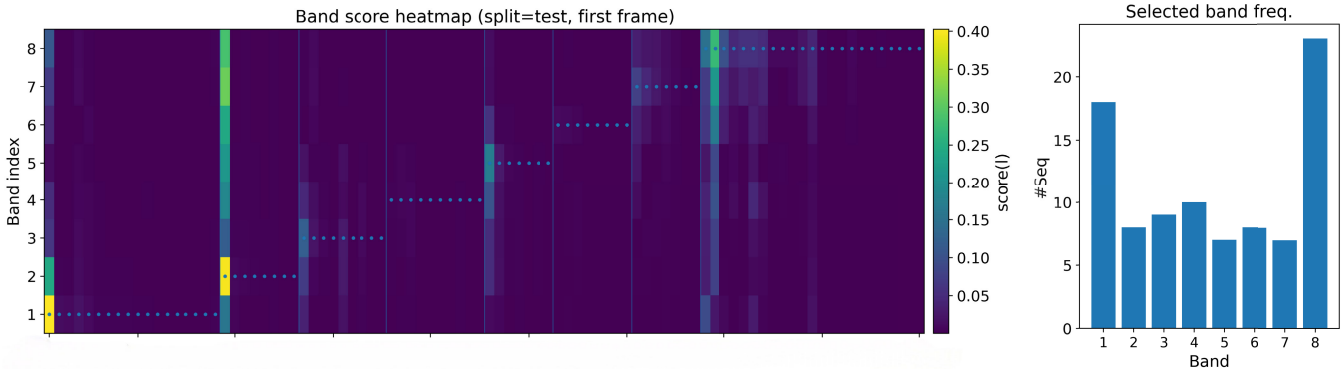


Fig. 8. Per-sequence band scores and the distribution of selected bands on the MUST test set.

(Bands 2/3/5); 2) a single NIR band (Band 8); 3) our target-aware band selection; and 4) multiband fusion.

Table XII shows that all variants outperform UNTrack*, confirming the benefit of introducing spectral observation evidence into the history construction. Band 8 alone already yields strong gains (AUC 65.76, PreN 82.61), which is consistent with the fact that NIR is often less sensitive to visible-texture clutter. Our band selection performs best, while selecting only *one* band from the first frame and keeping it fixed throughout the sequence, thereby preserving train–test consistency with negligible extra overhead. Fig. 8 further reveals substantial band-wise variability across sequences, explaining why fixed three-band fusion can underperform despite using more channels: mixing less informative bands

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TABLE XII

ABLATION OF BAND SELECTION STRATEGIES FOR HISTORICAL OBSERVATIONS ON MUST

Method	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
UNTrack*	62.75	78.58	53.32	82.46	77.50
3-band fusion (2/3/5)	65.25	81.82	57.70	84.36	82.54
NIR (Band 8)	65.76	82.34	57.96	84.88	82.61
Band selection (ours)	**66.56**	**83.49**	**58.55**	**86.07**	**83.82**
Multi-band fusion	66.06	83.03	58.22	85.31	82.80

TABLE XIII

ABLATION ON THE NUMBER OF HISTORICAL FRAMES ON MUST

History length	AUC	$SR_{0.5}$	$SR_{0.75}$	Pre	PreN
0 (UNTrack*)	62.75	78.58	53.32	82.46	77.50
1	65.03	81.50	57.23	83.93	81.44
2	**66.56**	**83.49**	**58.55**	**86.07**	**83.82**
3	65.38	82.27	56.20	84.99	82.17

may dilute discriminative cues and propagate biased history. Multiband fusion is also competitive (AUC 66.06, Pre 85.31), but likely suffers from redundant or noisy channels, suggesting reliability-aware band weighting as a promising direction for future work. This result suggests that reliable historical evidence depends more on target–background separability and background stability than on simply increasing the number of spectral channels.

9) Ablation on the Number of Historical Frames: Although different history lengths determine the amount of available temporal state information, and a longer temporal window can, in principle, capture higher-order dynamics, our historical evidence at the inference stage is derived from imperfect predictions. Therefore, incorporating too many past frames may amplify biased information and lead to error accumulation. To determine an appropriate history length, we conduct an ablation experiment on the short-term history window used in the HEI module.

Table XIII shows that even one historical frame already brings clear gains over UNTrack* (AUC: 62.75 → 65.03; $SR_{0.75}$: 53.32 → 57.23), indicating that the most recent state provides an effective prior, especially for strict-overlap quality. Increasing the history length to two frames yields the best overall performance (AUC 66.56, PreN 83.82), suggesting that two-frame history provides not only the latest state but also short-term temporal trends that improve scale-normalized localization accuracy. Extending the history to three frames degrades performance (AUC 65.38; $SR_{0.75}$ 56.20), which is consistent with error accumulation outweighing the benefit of higher-order cues. We therefore use two historical frames by default as the best tradeoff between short-term dynamics modeling and limiting error accumulation.

To further examine temporal behavior, Fig. 9 reports sequence-level IoU and normalized center location error (NCLE) curves with interquartile ranges. Hist = 2 maintains higher IoU and slower NCLE growth in the middle and late stages, further supporting the tradeoff between short-term trend modeling and error accumulation. Hist = 1 lacks sufficient temporal trend cues and therefore degrades faster under occlusion and viewpoint changes, while Hist = 3 may introduce stale or biased evidence under rapid moving-platform motion. This explains why a compact two-frame window is preferable to either single-frame history or longer history accumulation.

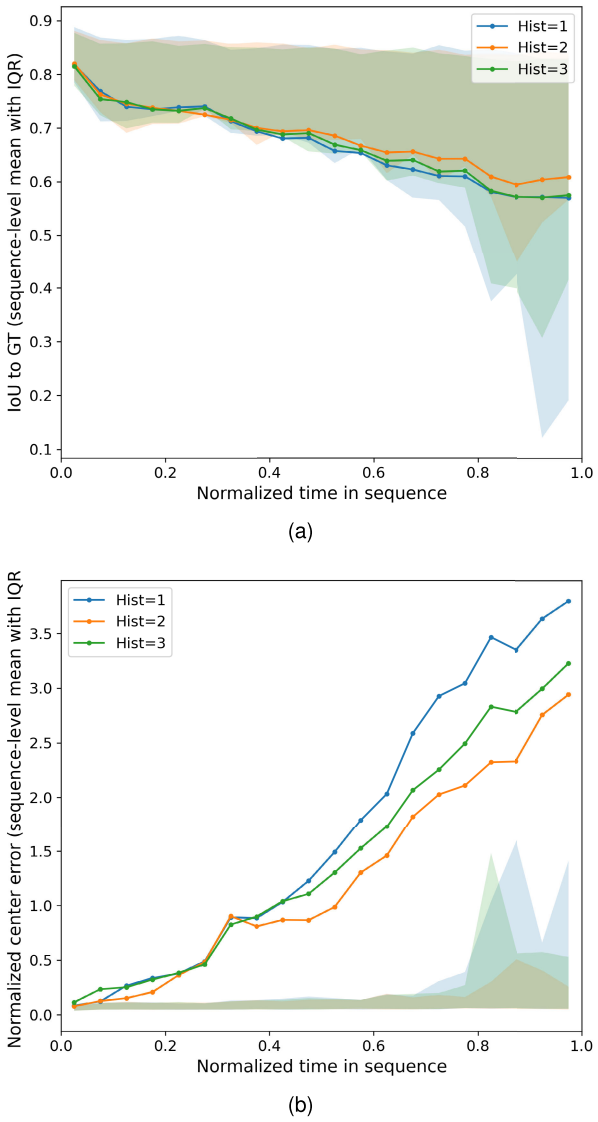


Fig. 9. Temporal behavior under different history lengths on MUST. (a) IoU to ground truth with sequence-level mean and interquartile range. (b) NCLE with sequence-level mean and interquartile range.

F. Limitations

Although HEATTrack balances short-term temporal evidence and anchor-guided target identity cues, it still depends on sufficiently reliable historical observations and template information. When the target undergoes long-term disappearance, severe occlusion, or persistent ambiguity with nearby distractors, the constructed evidence maps may become weak or biased because recent observations no longer provide reliable target information. In such cases, short-term historical evidence can help stabilize tracking only when the recent target state remains partially observable.

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In addition, auxiliary templates can still be contaminated by tracking noise, which may weaken the target-aware prompt despite the anchor-biased reliability fusion. Moreover, TAMTP provides an anchor-guided identity cue rather than a full long-horizon memory mechanism, and the first-frame anchor may become less representative under extreme appearance changes. Another limitation is that the selected historical observation band is determined from the first frame and kept fixed throughout the sequence. This design is efficient and avoids unstable online band switching, but it may become suboptimal under great spectral changes, illumination shifts, or scene-dependent band noise. Representative failure cases are provided in Appendix D.

V. Conclusion

We presented HEATTrack, a stability-oriented framework for aerial multispectral SOT. The main finding of this work is that stable aerial multispectral tracking benefits from a reliability-aware balance between short-term temporal evidence and long-term target identity cues. HEI provides lightweight recent-frame evidence for temporal adaptability, while TAMTP preserves anchor-guided identity cues for distractor resistance and identity discrimination. Experiments on MUST and MSITrack demonstrate that these two sources are complementary, improving both overlap quality and scale-normalized localization accuracy over strong baselines. Efficiency analysis further shows that HEI introduces only marginal overhead, whereas the main runtime cost comes from TAMTP; nevertheless, the full model remains more efficient than the original UNTrack Search = 2 protocol in terms of FLOPs. Future work will explore lighter prompt designs, reliability-aware multiband history modeling, and stronger long-horizon robustness under severe occlusion scenarios.

Appendix: Additional Implementation Details and Analyses

This appendix provides additional implementation details and supplementary analyses, including: 1) the reproduction configuration of UNTrack*; 2) the sequence-wise band selection used in HEI; 3) the design of the TAMTP module and its default hyperparameters; and 4) representative failure cases that illustrate the practical limitations of HEATTrack.

A. Reproduction Configuration of UNTrack*

To improve reproducibility and clarify the reproduced baseline used in our comparisons, we summarize the configuration of UNTrack* in Table XIV. Unless otherwise specified, UNTrack* follows the original UNTrack input protocol and uses two search frames. We follow the original UNTrack configuration as closely as possible; the main adjustment is the initial learning rate used for stable reproduction under our environment.

Upon publication, we will release the source code, trained models, training and testing scripts, result files, and environment dependencies to facilitate reproducibility.

TABLE XIV
Reproduction Configuration of UNTrack*

Item	Setting
Backbone	ViT-B/16
Search protocol	Search=2 unless otherwise specified
Search size	384×384
Template size	192×192
Training epochs	50
Effective batch size	16
Optimizer	AdamW
Initial learning rate	4×10^{-4}
LR decay	$\times 0.1$ after epoch 30
Training split	Official MUST/MSITrack split
External data	None
Loss function	Same as UNTrack

B. Sequence-Wise Band Selection

We compute the sequence-wise evidence band once from the first-frame target box and reuse it throughout the sequence. Given a target box $b = (x, y, w, h)$, we first convert it to the corner form (x_0, y_0, x_1, y_1) and construct two local regions for band scoring.

1) Foreground Region: The target box is shrunk around its center by $\text{shrink}_{fg} = 0.95$ to form the foreground region. This shrink operation reduces boundary contamination from nearby background pixels, which is important because aerial targets are often tiny and the annotated box may contain a non-negligible amount of surrounding background.

2) Local Background Region: We expand the target box by $\text{ring}_{scale} = 2.0$ to form an outer box, and define the local background as the ring region between the target box and the expanded box. This local background is used instead of the whole image background because nearby distractors and local clutter are more relevant to tracking failure than far-field regions.

For each band, we compute the foreground/background mean and variance over the two regions, and select the band according to the Fisher-separability and background-stability score defined in (4)–(7) of the main article, with $\epsilon = 10^{-6}$ and $\beta = 1$ by default. If either the foreground or background region is invalid, the procedure falls back to a valid default band for that sequence.

C. TAMTP Module

This section summarizes the implementation details of the TAMTP module. All mathematical definitions follow Section III-B of the main article, including the appearance prototype, structural embedding, spectral embedding, prompt injection, and reliability-aware template fusion.

1) Mask Alignment and Local Background Construction: For each template, the pixel-level target box mask is aligned to different feature grids according to the branch where it is used. For the appearance prototype, the mask is downsampled to the template-token grid using area-preserving average pooling. For the structural and spectral branches, the mask is aligned to the wavelet-map resolution; we use area-preserving pooling

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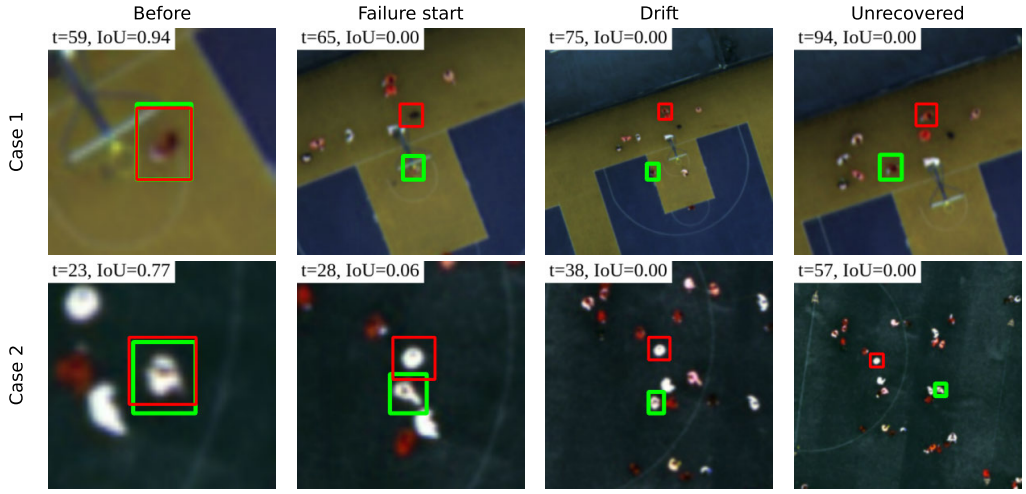


Fig. 10. Representative failure cases of HEATTrack. Red and green boxes denote our prediction and the ground truth, respectively. The four columns show the tracking process before failure, at failure onset, during drift, and after persistent nonrecovery. Although HEATTrack can accurately localize the target before failure, it may still suffer from persistent drift when the target becomes extremely small, ambiguous, or confused with nearby distractors. The IoU values above each crop indicate that the tracker fails to recover after the failure onset.

when the spatial size is divisible and bilinear interpolation otherwise. The local background region is constructed as a ring around the target on the corresponding grid, following Section III-B. This local construction reduces the influence of far-field background statistics and focuses the prompt on nearby distractors that are more likely to cause tracking failure.

2) Wavelet-Based Structural and Spectral Cues: For the structural branch, we apply a single-level 2-D Haar transform to each template and aggregate the magnitudes of the three high-frequency sub-bands, i.e., LH, HL, and HH. The magnitude aggregation reduces sensitivity to small shifts and coefficient sign changes while preserving complementary directional structures. For the spectral branch, we first apply per-pixel ℓ_1 normalization along the band dimension to suppress global intensity variation. We then perform a single-level 1-D Haar transform along the wavelength dimension and concatenate the low-frequency trend and high-frequency details as the spectral-shape map. Both branches append the aligned ROI mask as an explicit spatial prior.

3) Lightweight Encoders: The structural encoder Φ_e and spectral encoder Φ_s share the same lightweight architecture. Each encoder takes a nine-channel input, consisting of an 8-channel wavelet feature map and a 1-channel aligned ROI mask, and outputs a 128-channel feature map at the wavelet resolution. The default architecture is:

1) 1×1 Conv $(9 \rightarrow 128)$+ GroupNorm + SiLU;
2) depthwise 3×3 Conv $(128 \rightarrow 128)$+ GroupNorm + SiLU; and
3) 1×1 Conv $(128 \rightarrow 128)$+ GroupNorm + SiLU.

GroupNorm uses up to 32 groups, selecting the largest divisor of the channel number that does not exceed 32.

4) Prompt Injection and Reliability Fusion: The structural and spectral embeddings are projected to the prompt channel dimension and injected into the appearance prototype through LayerScale residuals. The LayerScale parameters are initialized as $\alpha_e = \alpha_s = 10^{-3}$, so that the module starts close to an appearance-only prompt and gradually learns to use structural and spectral residuals during training. For multitemplate fusion, we use the first-frame template as the anchor because its box annotation is reliable. The reliability score of each auxiliary template is computed from its cue consistency with the anchor template, and the final fusion weights are obtained by a softmax with an anchor bias $\delta = 1.0$. This anchor-biased design provides a fallback when auxiliary templates are contaminated by tracking noise, box misalignment, or nearby distractors.

5) Template Setting and Training-Time Jitter: We use $K = 3$ templates in all experiments. During training, auxiliary template boxes are jittered using the same perturbation-and-annealing schedule as HEI to simulate inference-time template misalignment. The first-frame anchor template is kept unjittered. Unless otherwise stated, we use $B = 8$, $d_e = d_s = 128$, ctx_kernel $= 5$, enable per-pixel spectral normalization and reliability detachment, set $\delta = 1.0$, initialize $\alpha_e = \alpha_s = 10^{-3}$, and disable the postfusion MLP. Reliability weights are computed on detached representations for training stability.

D. Additional Qualitative Analysis

Fig. 10 shows representative failure cases of HEATTrack. Although the tracker localizes the target accurately before failure, it may still suffer from persistent drift when the target becomes extremely small, visually ambiguous, or confused with nearby distractors. After the failure onset, the predictions remain far from the ground truth for a long duration, indicating that short-term historical evidence and auxiliary templates may become unreliable in such cases.

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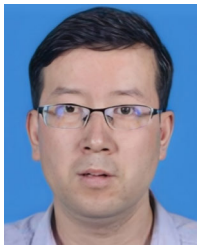
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